

In parametric vector form, the solution set is:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -9x_2 + 4x_3 - 2 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -9x_2 \\ x_2 \\ 0 \end{bmatrix} + \begin{bmatrix} 4x_3 \\ 0 \\ x_3 \end{bmatrix} + \begin{bmatrix} -2 \\ 0 \\ 0 \end{bmatrix} \quad (\text{Vector addition})$$

$$= x_2 \begin{bmatrix} -9 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 4 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} -2 \\ 0 \\ 0 \end{bmatrix} \quad (\text{Scalar multiplication})$$

The solution is the plane through the line $\begin{bmatrix} -2 \\ 0 \\ 0 \end{bmatrix}$ parallel to the plane spanned by the vectors $\begin{bmatrix} -9 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 4 \\ 0 \\ 1 \end{bmatrix}$. The plane spanned by

the given vectors is $x_1 + 9x_2 - 4x_3 = 0$ (By Theorem 6) $\square$

19. Find the parametric equation of the line through a parallel to b .

$$a = \begin{bmatrix} -2 \\ 0 \end{bmatrix}, b = \begin{bmatrix} -5 \\ 3 \end{bmatrix} \quad x = a + tb$$

Solution:

$$\begin{bmatrix} -2 \\ 0 \end{bmatrix} + t \begin{bmatrix} -5 \\ 3 \end{bmatrix} = x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$x_1 = -5t - 2$$

$$x_2 = 3t$$

21. Find a parametric equation of the line M through P and Q .

$$P = \begin{bmatrix} 2 \\ -5 \end{bmatrix}, Q = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

Hint: See the figure. M is parallel to the vector $(Q-P)$.

The line through P and Q

$$Q - P = \begin{bmatrix} -5 \\ 6 \end{bmatrix} = t \quad x = P + t(Q - P) \Rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ -5 \end{bmatrix} + t \begin{bmatrix} -5 \\ 6 \end{bmatrix}$$

$$\Rightarrow x_1 = 2 - 5t \text{ and } x_2 = -5 + 6t$$